CHAPTER

INTRODUCTION

Prefix	Factor	Symbol
deci	10^{-1}	d
centi	10^{-2}	c
milli	10^{-3}	m
micro	10^{-6}	μ
nano	10^{-9}	n
pico	10^{-12}	p
femto	10^{-15}	f
atto	10^{-18}	a

deca	10^{1}	da
hecto	10^{2}	h
kilo	10^{3}	K
mega	10^{6}	M
giga	10^{9}	G
tera	10^{12}	T
peta	10^{15}	P
exa	10^{18}	E

Date

Numbers

Exact — no change

Inexact — varies

SIGNIFICANT FIGURES

The number of significant figures is the number of digits believe to be ~~exact~~ correct by the person doing the measurement.

OR

The sum of certain or uncertain is called SF

OR

The digits whose values are accruatly known in a particular measurement are called its SF

EXAMPLE

1. 0024004100

extreme left — middle — extreme right

2. $\underline{22}\overline{9}$ — uc, c — 1+2 = 3 SF

3. $\underline{53}\overline{0}$ — uc, c — 1+1 = 2

529.9

4. 0.0068

5. 1.073

6. 300

7. 300.0

8. 4.57x10

9. 300.

RULE

4. 0.00683 3 SF

5. 1.073 4 SF

6. 300 1 SF

7. 300.0 4~~3~~ SF

8. 4.57×10^8 3 SF

9. 300. 3 SF

RULES

ROUND OFF RULES

* if the number is before 5 is even, then remains same.

78.5 ⇒ 78

* if the number before the 5 is odd then add 1

77.5 = 78

Date

Q) Give answers to the correct number of SF

1) 4503 + 34.9 + 550

$$\begin{array}{r} 4503 \\ 550 \\ +\ 34.9 \\ \hline 5087.9 \\ \hline \end{array}$$

= 5087.9

= 5088

Answer

Q) Which of the following length measurement is more accurate.

* 500.0cm — 4 SF
* 0.005 cm — 1 SF
* 5.00cm — 3 SF

→ 500.0 cm is more accurate then 0.005cm and 50.00 cm bcuz it has larger number of SF.

Q) Difference b/w 2.0 and 2.000

Q) In a simple pendulum experiment the length of the pendulum is 90.6×10^{-2} m. The time period is 1.91 sec. Write the values of acceleration due to gravity to correct S.F and round off

$L = 90.6 \times 10^{-2} m$

$T = 1.91$ sec

$g = ?$

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$1.91 = 2\pi \sqrt{\frac{90.6 \times 10^{-2}}{g}}$$

$$g = \frac{4\pi (90.6 \times 10^{-2})}{(1.91)^2}$$

$$g = 9.806 \, m/s^2$$

round off;

$$g = 9.8 \, m/s^2$$

$$g = 10 \, m/s^2$$

ERROR

The deviation of the measured value with the actual value is known as error

Error can be categorized into

Random Error and Systematic Error

Random Error

- It is due to unknown causes
- varies about true and actual values
- due to unknown causes we can't depend upon a single observation must be repeated atleast 3 times. It can be removed by mean and average method.

Systematic Error

- It is the error due to non causes contributes to the contant values and measured values
- It can be eliminated by suitable method.
- Since we know the cause it can't be eliminated by mean method.

COMBI
RAND

if q is
resultant,

Δq; e
Δa; er
Δb; er

ADDI

$(q \pm$
$(q \pm$
$q \pm$
$\pm \Delta$

Combination Effect On Random Error

if q is the actual or true vale of the resultant, a and b are the true vale of the varibles

$$q = a + b \quad \text{.....} \ (1)$$

Δq ; error in the measurement of q,

Δa ; error in the measurement of a

Δb ; error in the measurement of b

Addition

$$(q \pm \Delta q) = (a \pm \Delta a) \pm (b \pm \Delta b)$$

$$(q \pm \Delta q) = (a + b) \pm (\Delta a + \Delta b)$$

$$q \pm \Delta q = q \pm (\Delta a \pm \Delta b)$$

$$\pm \Delta q = \pm (\Delta a + \Delta b)$$

$$\Delta q = \Delta a + \Delta b$$

SCIENTIFIC NOTATION

PREFIX	FACTOR	SYMBOL
deci	10^{-1}	d
centi	10^{-2}	c
milli	10^{-3}	m
micro	10^{-6}	μ
nano	10^{-9}	n
pico	10^{-12}	p
femto	10^{-15}	f
atto	10^{-18}	a

PREFIX	FACTOR	SYMBOL
deca	10^1	da
hecto	10^2	h
killo	10^3	K
mega	10^6	M
giga	10^9	G
tera	10^{12}	T
peta	10^{15}	P
exa	10^{18}	E

Numbers can be divided into two categories;

Exact
no change

Inexact
variation / changing

SIGNIFICANT FIGURES

The sum of certain and uncertain is called significant figures.

example

$\underline{00}24\underline{00}4\underline{00}$

extrem left ← ↓ middle → extreme right

$\underset{c}{2}\overset{uc}{\overline{29}}$ SF = 1+2 = 3

$\underset{c}{5}\overset{uc}{\overline{30}}$ SF = 1+1 = 2

529.9

The digits whose values are accurately in a particular measurement are called SF

i. Leading zero's are never significant. All non zero's digits are significant.
ii. All zero's occuring b/w two non zero's digits are significant.
iii All zero's to the right of decimal point and to the left of non-zero are never significant.
iv. Embeded zero's are significant.
v. Trailing zero are non-significant
vi Trailing zero are significant only if decimal point is ~~eqpt~~ specified

*) These rules are very complicated

Just rewise from XI (Sir Zain Notes) For S.F and all operations.

Question

Write down the number of SF 6729 ..

i	6729	4
ii	0.024	2
iii	6.0023	5
iv	2.520×10^7	4
v	2.520	4
vi	0.08240	4
vii	4200	2
viii	4.57×10^8	3
ix	91.000	5

*

ROUND OFF

→ if the no. before 5 is even
let it be

→ if the no. after after 5 is odd
round off
/
add 1

→ if no. is 5
↳ check two conditions

Question

Which of
more accu

i 500.0

iii 5.00 c

i) 4 ~~51~~

500.0 c
larger
Question

Differenc

Question

Which of the following length measurement is more accurate?

i 500.0 cm   ii 0.005 cm

iii 5.00 cm

i) 4 SF   ii) 1 SF   iii) 3 SF

500.0 cm is more accurate bcuz it has larger no. of SF.

Question

Difference b/w 2.0 and 2.000

$$\begin{array}{r} 2.000 \\ -\,2.0 \\ \hline 0.000 \end{array}$$

Question

In a simple pendulum experiment the length of string is 90.6cm and time period is 1.91 sec. Write the acceleration due to gravity to correct SF and round off

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$\frac{T^2}{4\pi^2} = \frac{l}{g} \qquad g = \frac{4\pi^2 l}{T^2}$$

$$g = \frac{4\pi^2 (90.6 \times 10^{-2})}{(1.91)^2}$$

$$g = \cancel{9.804} - \cancel{9.8}$$

$$g = 9.806 \, m/sec^2$$

$$g = 10 \, m/sec^2$$

ERROR

The deviation in the measure value and the actual value is called error

Can be broadly classified into 2 types;

Random Error

→ It is due to unknown causes. Random error can change into polarity.

→ Varies about true and actual values

→ due to unknown causes we can't depend upon a single observation must be repeated atleast 3 times. It can be removed by mean and average method

Systematic

It is the error due to known causes, contributes to the constant values and measured values. systematic error can be eliminated by suitable method. Since we know the cause it can't be eliminated by mean method.

EFFECT OF COMBINING ERROR

IF q is the actual or true value of the resultant, a and b are the true value of the parameters /variables

$$Q = a + b$$

Δq ; is the error in the measurement of q

Δa ; is the error in the measurement of a

Δb ; is the error in the measurement of b

Addition

$$(q \pm \Delta q) = (a \pm \Delta a) \pm (b \pm \Delta b)$$

$$(q \pm \Delta q) = (a + b) \pm (\Delta a \pm \Delta b)$$

$$q \pm \Delta q = q \pm (\Delta a \pm \Delta b)$$

$$\pm \Delta q = \pm (\Delta a + \Delta b)$$

$$\Delta q = \Delta a + \Delta b$$

$$\therefore \Delta q = \Delta a + \Delta b$$

Subtraction

$$q = a - b$$

$$(q \pm \Delta q) = (a \pm \Delta a) - (b \pm \Delta b)$$

$$(q \pm \Delta q) = (a - b) \pm (\Delta a + \Delta b)$$

$$q \pm \Delta q = q \pm (\Delta a + \Delta b)$$

$$\Delta q = \Delta a + \Delta b$$

$$\therefore \Delta q = \Delta a + \Delta b$$

Two quantities are added or subtracted, the ~~result~~ absolute error and the final result is the sum of the absolute error in the quantities.

Multiplication

$$q = a \cdot b \rightarrow (i)$$

$$q \pm \Delta q = (a \pm \Delta a) \cdot (b \pm \Delta b)$$

$$q \pm \Delta q = (a \cdot b) \pm a\Delta b \pm b\Delta a \pm \underbrace{\Delta a \Delta b}_{\text{neg}}$$

$$q \pm \Delta q = (a \cdot b) \pm a\Delta b \pm b\Delta a \rightarrow (ii)$$

$\div$ing eq (ii) by (i)

$$\frac{q \pm \Delta q}{q} = \frac{(a \cdot b) \pm a\Delta b \pm b\Delta a}{a \cdot b}$$

$$\frac{q \pm \Delta q}{q} = \frac{a \cdot b}{a \cdot b} \pm \frac{a\Delta b}{a \cdot b} \pm \frac{b\Delta a}{a \cdot b}$$

$$1 \pm \frac{\Delta q}{q} = 1 \pm \frac{\Delta b}{b} \pm \frac{\Delta a}{a}$$

$$\frac{\Delta q}{q} = \pm \frac{\Delta a}{a} \pm \frac{\Delta b}{b}.$$

If q is the product of two quantities a and b the fractional error in q is the sum of the fractional error in a and b.

Question

If the capacitance of a capacitor is $(2 \pm 0.1)F$ and the potential V is (25 ± 0.25). Find the error in the change of q. Find the error in q (charge)

$$q = C \times V$$

$$q = 2 \times 25 = 50\ C$$

$$\frac{\Delta q}{q} = \frac{\Delta C}{C} + \frac{\Delta V}{V}$$

$$\frac{\Delta q}{50} = \frac{0.1}{2} + \frac{0.25}{25}$$

$$\Delta q = 3\ C$$

instead of -ve we use +ve b/c we always calculate max. value

Division

if $Q = \frac{a}{b}$ $\Rightarrow Q = ab^{-1}$ $\rightarrow$ eq 1

$$Q \pm \Delta Q = (a \pm \Delta a)(b \pm \Delta b)^{-1}$$

$$= (a \pm \Delta a)(b^{-1} \pm b^{-2}\Delta b \pm \dots)$$

$$= (a \pm \Delta a)(b^{-1} \pm b^{-2}\Delta b)$$

$$Q \pm \Delta Q = ab^{-1} \pm ab^{-2}\Delta b \pm b^{-1}\Delta a \pm \underset{\text{negligible}}{b^{-2}\Delta b\,\Delta a}$$

$$Q \pm \Delta Q = ab^{-1} \pm ab^{-2}\Delta b \pm b^{-1}\Delta a \rightarrow \text{eq } 2$$

÷ing eq 2 by eq 1

$$\frac{Q \pm \Delta Q}{Q} = \frac{\cancel{ab^{-1}}}{\cancel{ab^{-1}}} \pm \frac{ab^{-2}\Delta b}{ab^{-1}} \pm \frac{b^{-1}\Delta a}{ab^{-1}}$$

$$\cancel{1} \pm \frac{\Delta Q}{Q} = \cancel{1} \pm \frac{\Delta b}{b} \pm \frac{\Delta a}{a}$$

$$\frac{\Delta Q}{Q} = \pm\left(\frac{\Delta a}{a} \pm \frac{\Delta b}{b}\right)$$

Conclusion

Two quantities are multiplied or divide then absolute error and the final result is the sum of the fractional error in the quantities.

ERROR CALCULATION WITH POWER

$$Q = a^n$$

$$\frac{\Delta Q}{Q} = n \frac{\Delta a}{a}$$

more than one value:

$$Q = K a^x b^y c^z$$

$$Q = K \left[x \frac{\Delta a}{a} + y \frac{\Delta b}{b} + z \frac{\Delta c}{c} \right]$$

Question

The length of the square is measured 10 cm and the measurement has error of ±0.5 cm. Find the absolute error.

Solution

Let, length $= x = 10$ cm

then,. Area $= x^2 = (10)^2 = 100$ cm^2

$$\because A = x^2$$

$$A = L \times L = 10 \times 10 = 100 cm^2$$

$$\therefore \frac{\Delta A}{A} = 2 \frac{\Delta x}{x}$$

$$\frac{\Delta A}{A} = 2 \times \frac{0.5}{10} \quad 2. \quad \frac{}{10}$$

$$\frac{2.0.5}{10} \cdot 10.10$$

$$\Delta A = 0.1 \times 100$$

$$\Delta A = 10 \, cm^2$$

$$A \pm \Delta A = (100 \pm 10) \, cm^2$$

Question

Mass of an object is (345.1 ± 0.1) gm and the volume is (41.55 ± 0.05) cm^3. Calculate the error in density.

$$\because \rho = \frac{m}{V}$$

$$\rho = \frac{345.1}{41.55} = 8.305 \ g/cm^3$$

$$\because \frac{\Delta\rho}{\rho} = \frac{dm}{m} + \frac{\Delta V}{V}$$

$$\frac{\Delta\rho}{\rho} = \frac{0.1}{345.1} + \frac{0.05}{41.55}$$

$$\frac{\Delta\rho}{\rho} = 1.49 \times 10^{-3} \ g/cm^3$$

$$\Delta\rho = 1.49 \times 10^{-3} \times 8.305$$

$$\Delta\rho = 0.01 \ g/cm^3$$

$$(\rho \pm \Delta\rho) = (8.305 \pm 0.01) \ g/cm^3$$

Question

In an experiment, determine e/m ratio using cylindrical diode.
The following eq. defined as:

$$\frac{e}{m} = \frac{8V}{r^2B^2}$$

where, V is (20 ± 1) volt
$r = (3.4 \pm 0.1)$ mm
$B = (9.08 \pm 0.09)$ mT

Determine the absolute error in $\frac{e}{m}$

$r = (3.4 \pm 0.1) \times 10^{-3}$ m
$B = (9.08 \pm 0.09) \times 10^{-3}$ T
$B = 9.08$ mT $= 9.08 \times 10^{-3}$ T

$$\frac{e}{m} = \frac{8 \times 20}{8\,(3.4\times10^{-3})^2(9.08\times10^{-3})^2}$$

$$\frac{e}{m} = (3.4\times10^{-3})^2(9.08\times10^{-3})^2$$

$$\frac{e}{m} = 1.67\times10^{11}\ C/Kg$$

$$\frac{e}{m} = 1.67\times10^{11}\ c/kg$$

Now,

$$\frac{\Delta e/m}{e/m} = \frac{\Delta V}{V} + 2\frac{\Delta r}{r} + 2\frac{\Delta B}{B}$$

$$\frac{\Delta e/m}{e/m} = \frac{1}{2020} + 2\left(\frac{0.1\times10^{-3}}{3.4\times10^{-3}}\right) + 2\left(\frac{0.09\times10^{-3}}{9.08\times10^{-3}}\right)$$

$$\frac{\Delta e/m}{e/m} = 0.05 + 0.05 + 0.019$$

$$\frac{\Delta e/m}{e/m} = 0.128$$

$$\Delta e/m = 0.128 \times 1.67\times10^{11}$$

$$\Delta e/m = +0.213\times10^{11}\ \mathrm{c/kg}$$

$$e/m \pm \Delta e/m = (1.67 \pm 0.213)\times10^{11}\ \mathrm{c/kg}$$

Δx = absolute error

$\frac{\Delta x}{x}$ = relative error

$\frac{\Delta x}{x} \times 100$ = %age error

TOTAL DIFFERENTIATION

$$z = f(x, y)$$

$$dz = \frac{dz}{dx} \Delta x + \frac{\partial z}{\partial y} \Delta y$$

↳ approximation formula / combine effect of error

$Q = CV \Rightarrow Q = xy$

$$dQ = \frac{\partial Q}{\partial x} \Delta x + \frac{\partial Q}{\partial y} \Delta y$$

$$dQ = y \Delta x + x \Delta y$$

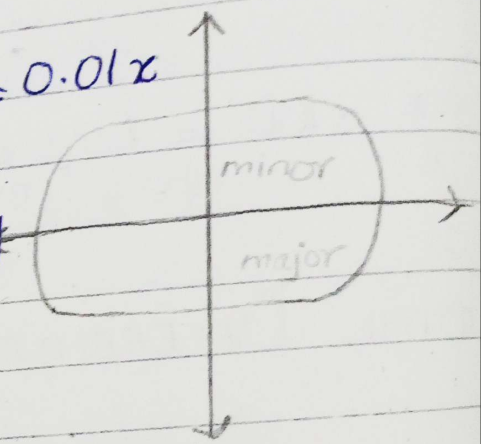
v.v.v Imp.

~~Area of an ellipse~~

Question

Find the percentage error in the area of an ellipse if 1% error is made in measuring the major and minor axis.

$\Delta x = 1\%$ of $x = \frac{1}{100} = 0.01x$



$\Delta y = 1\%$ of $y = \frac{1}{100} = 0.01y$

Area of ellipse $= \pi xy$

$$\partial A = \frac{\partial A}{\partial x} \Delta x + \frac{\partial A}{\partial y} \Delta y$$

$$\partial A = \pi y \Delta x + \pi x \Delta y$$

$$\partial A = \pi y (0.01x) + \pi x (0.01y)$$

$$\partial A = 0.01 xy\pi + 0.01 xy\pi$$

$$\partial A = 0.02 xy\pi$$

$$\partial A = 0.02 (A)$$

Now,

$$\% \partial A = 0.02 \times 100 \; (A)$$

$$\% \partial A = 2A$$

%age error in the area of an ellipse
$= 2\%$ of A

Question

The time period of simple pendulum is $T = 2\pi\sqrt{L/g}$. Find the maximum error in time period. Due to possible error upto 1 % in length 'L' and 2% in acceleration due to gravity 'g'

$$\%\text{age error in } L = 100 \times \frac{\Delta L}{L} = 1$$

$$\%\text{age error in } g = 100 \times \frac{\Delta g}{g} = 2$$

$$T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow T = 2\pi\left(\frac{L}{g}\right)^{1/2}$$

Taking log on both side

$$\log T = \log 2\pi + \frac{1}{2}\log\left(\frac{L}{g}\right)$$

$$\log T = 0 + \frac{1}{2}\frac{\Delta L}{L} - \frac{1}{2}\frac{\Delta g}{g}$$

If $\times 100 \frac{\Delta T}{T} = \frac{1}{2}\left(100 \times \frac{\Delta L}{L} - 100 \times \frac{\Delta g}{g}\right)$

the

fra

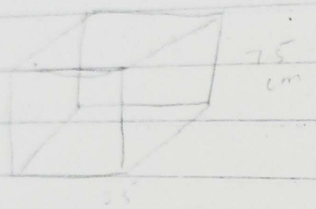
$$100 \times \frac{\Delta T}{T} = \frac{1}{2}\left(1 \pm 2\right)$$

$$100 \times \frac{\Delta T}{T} = \frac{1}{2}(3)$$

max. %age error in T = 1.5%.

Question

A side of a cube is measured (7.5 ± 0.1) cm length cube. Find the change in volume



$V = a^3 = (7.5)^3 = V = 421.87$

FORMULA OF APPROXIMATION.

$$\frac{\Delta V}{V} = 3\frac{\Delta a}{a}$$

$$\frac{\Delta V}{V} = \left(\frac{3 \times 0.1}{7.5}\right) \times 421.875$$

$$\frac{\Delta V}{V} = 16.875 \approx 17\ cm^3$$

Question

The force
mass
a circle

The meas
$m = (3$
$V = ($
$r = (1$

Find the
and %age

$F \ \frac{\Delta F}{F}\ \frac{(3.5}{(}$

$F \ \frac{\Delta F}{F}\ 112N$

If
the
frac

$V = 17\ cm^3$

Question

Question

The force acting on an object of mass 'm' travelling at velocity 'v' in a circle of radius 'r' is given by

$$F = \frac{mv^2}{r}$$

The measurements are recorded as

$m = (3.5 \pm 0.1)\,kg$

$v = (20 \pm 1)\ m/s$

$r = (12.5 \pm 0.5)\,m$

$F = \frac{(3.5)(20)^2}{12.5} = 112\ N$

Find the max. possible fractional error and %age error in the force.

Now

$$F \quad \frac{\Delta F}{F} = \frac{(3.5)(20)^2}{(12.5)} \quad + 2\frac{\Delta v}{v} + \frac{\Delta r}{r}$$

$$F \quad \frac{\Delta F}{F} = 112N \quad \frac{0.1}{3.5} + 2\left(\frac{20}{1}\right) + \frac{0.5}{12.5}$$

Now,

$$\frac{\Delta F}{F} = \frac{\Delta m}{m} + 2\frac{\Delta v}{v} + \frac{\Delta r}{r}$$

$$\frac{\Delta F}{F} = \frac{0.1}{3.5} + 2\left(\frac{1}{20}\right) + \left(\frac{0.5}{12.5}\right)$$

$$\frac{\Delta F}{F} = 0.168 \approx 0.17$$

$$\frac{\Delta F}{F} \times 100 = 0.17 \times 100$$

%age error = 19.04 N

$F \pm \Delta F = (112 \pm 19)\,N$

Question

The power 'p' is decipated in resistor is given by $P = \frac{E^2}{R}$. If $E = 200V$ and $R = 8\,\Omega$. Find change in 'P' resulting from decrease of 5 volt in E and increase of 0.2 Ω in R

$E = (200 \pm 5)$ volts

$R = (8 \pm 0.2)\,\Omega$

$\Delta P = ?$

$$P = \frac{E^2}{R}$$

$$P = \frac{200^2}{8} \Rightarrow P = 5000\,\text{watt}$$

Now,

$$\frac{\Delta P}{P} = 2\frac{\Delta E}{E} + \frac{\Delta R}{R}$$

$$\Delta P = \left\{2\left(\frac{5}{200}\right) + \frac{0.2}{8}\right\} \times 5000$$

$\Delta P = \pm 375$ watt

$P \pm \Delta P = (5000 \pm 375)$ watt

Question

The length of the pendulum is 10 cm and is known 1 mm accuracy. The period of its oscillation is about 0.5 sec the time of 100 oscillation is measured with a watch of 1 sec rev. What is the accuracy of determination of g?

1 mm = 0.1 cm ; $\frac{(10 \pm 0.01)}{100}$ sec

$$T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow \frac{T^2}{4\pi^2} = \frac{L}{g} \Rightarrow g = \frac{4\pi^2 L}{T^2}$$

$$\Rightarrow g = \frac{4(3.142)^2(0.1)^2}{(0.5)^2}$$

$\Rightarrow g = 1579.13$

M Question

If $I = V/R$ and $V = 250$ volts, $R = 50\ \Omega$.
Find the change in I resulting from an increase of 1 volt in potential and increase of 0.5 in resistance.

$V = (250 \pm 1)$ volts

$R = (50 \pm 0.5)$ volts

$I \pm \Delta I = ?$

$$I = \frac{V}{R} = \frac{250}{50} = 5 \text{ Amp}$$

Now,

$$\frac{\pm \Delta I}{I} = \frac{\Delta V}{V} + \frac{\Delta R}{R}$$

$$\pm \Delta I = \left(\frac{1}{250} + \frac{0.5}{50} \right) \times 5$$

$$\pm \Delta I = 0.07$$

$$I \pm \Delta I = (5 \pm 0.07)\,\text{Amp}$$

M Question

The range of projectile "R" which starts with the velocity 'V' at an elevation is θ is given by

$$R = \frac{V^2 \sin 2\theta}{g}$$

Find the %age error in R due to an error of 1% V and 0.5% g.

$$\Delta g = 0$$

Solution

$$R = \frac{V^2 \sin 2\alpha}{g} \rightarrow (1)$$

$\frac{\partial g}{g}$

so,

$= \ln \frac{u^2 v}{w}$

$= \ln u^2 v - \ln w$

$= \ln u^2 + \ln v - \ln w$

$= 2 \ln u$

Taking log on b.s of 1

$\log R = 2 \log V + \log \sin 2\alpha - \log(g)$

diff we get

$$\frac{\partial R}{R} = \frac{2 \partial v}{v} + \frac{2 \cos 2\alpha \cdot \partial \alpha}{\sin 2\alpha} - \frac{\partial g}{g}$$

$$100 \cdot \frac{\partial R}{R} = 100 \cdot \frac{2\,\partial V}{V} + 100 \cdot \frac{2\cos 2\alpha \cdot \partial\alpha}{\sin 2\alpha} - 100\frac{\partial g}{g}$$

$$\cancel{= 2(1) + 2\cos 2\alpha}$$

$$= 2(1) + 2\alpha\cot 2\alpha\,(0.5)$$

$$= 2 + \alpha\cot 2\alpha$$

Solution

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = \frac{4\pi^2 L}{g}$$

$$g = \frac{4\pi^2 L}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta T}{T}$$

Here, $\Delta L = 1mm = 0.1cm$

$L = 10\ cm$

$T = 0.5 \times 100 = 50 sec$

$\Delta T = 1\ sec$

Now,

$$\frac{\Delta g}{g} = \frac{0.1}{10} + 2 \times \frac{1}{50} = 0.01 + 0.04$$

$$\frac{\Delta g}{g} = 0.05 \Rightarrow FE \text{ in } g$$

$$100 \times \frac{\Delta g}{g} = 0.05 \times 100 = 5\%$$

GRAPHS

Figure showing the relation b/w two or more quantities is known as graph.

TYPES OF GRAPH

- Plane Graph Paper
- Semi "
- Log to Log "
- Polar "

ADVANTAGES

There are two main advantages;

i) The variation of one quantity with another may be seen easily.

ii) The average value of a constant may be determined from the graph.

TYPES OF CURVE

i Straight Line → Linear curve
ii Non-Linear curve → Parabola, Hyperbola, Ellipse, circle

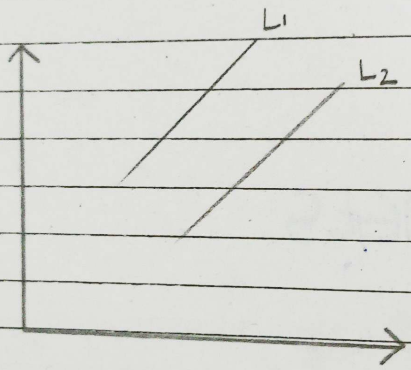
i) LINEAR CURVE

Linear curve only include the curve form by a straight line

$$y = mx + C$$

Both lines are represented by the equation of straight line

i.e $y = mx + C$



Same and +ve slope

Q) The $K.E = T = \frac{1}{2}mv^2$. Find the approximation change in total energy as mass changes from 45 to ~~45~~ 45.5 kg and velocity change from 1600 m/s to 1590 m/s.

$m = (45 \pm 45.5)\,kg$

$\Rightarrow m = (45 \times 10^{+3} \pm 45.5 \times 10^{3})\,g$

$v = (1600 \pm 1590)\,m/s$

K.E

$K.E = \frac{1}{2}$

K.E = 5

$\pm \frac{\Delta K.E}{K.E}$

$\pm \Delta K.E =$

$\pm \Delta K.E =$

$K.E \pm \Delta$

ii) NON

(a) Expon

$$K.E = \frac{1}{2} m v^2$$

$$K.E = \frac{1}{2} (45)(1600)^2$$

$$K.E = 57.6 \times 10^6$$

$$\pm \frac{\Delta K.E}{K.E} = \frac{\Delta m}{m} + \frac{2\Delta v}{v}$$

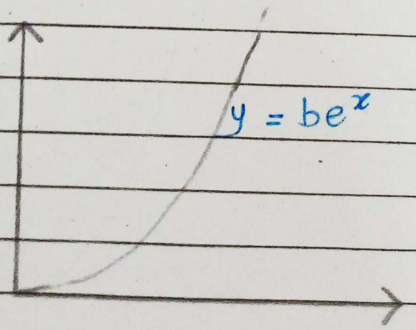
$$\pm \Delta K.E = \left\{ \frac{45.5}{45} + 2\left(\frac{1590}{1600}\right) \right\} \times 57.6 \times 10^6$$

$$\pm \Delta K.E = 172.72 \times 10^6 \ J$$

$$\boxed{K.E \pm \Delta K.E = (57.6 \pm 172.72) \times 10^6 \ J}$$

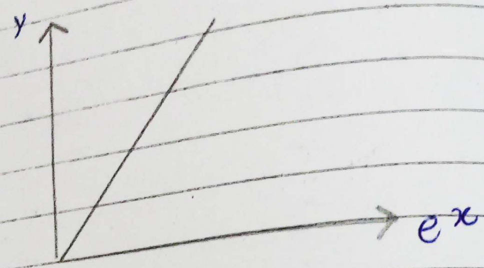
ii) Non - Linear Curve

(a) Exponentially Increasing

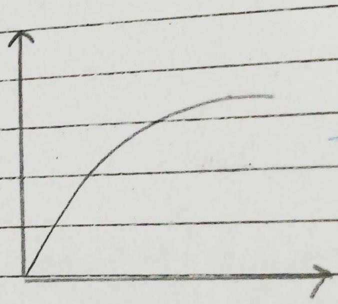


into linear curve

+ve increasing slope

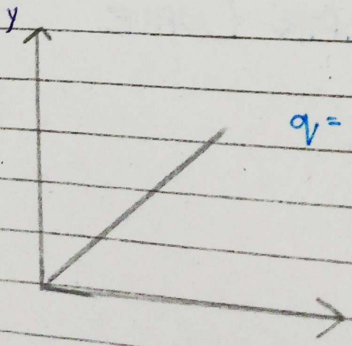


(b) Negative Curve



$y = K(1-e^x)$

into linear curve



$q = q_0(1-e^{-t/Rc})$

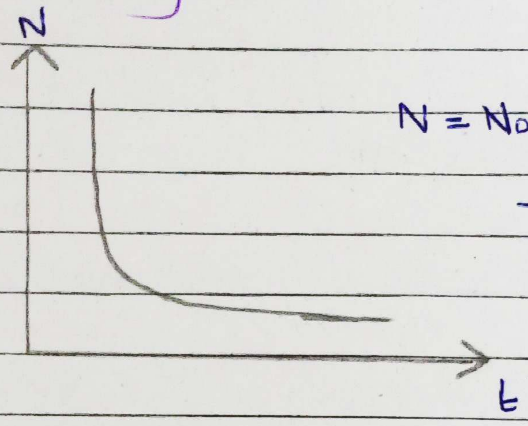
(c) Exponenti

into linear cur

(d) Parabo

into linear

(c):- Exponentially Curve

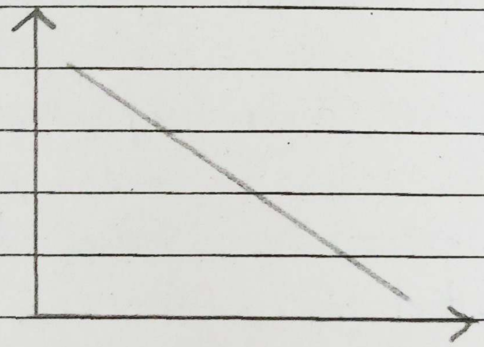


$N = N_0 Z^{-\lambda t}$

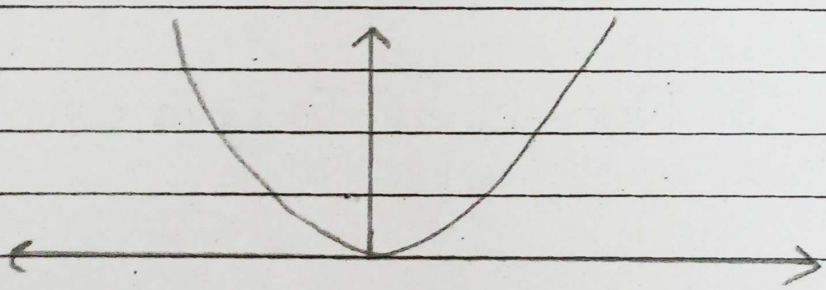
-ve decreasing slope

↓

into linear curve

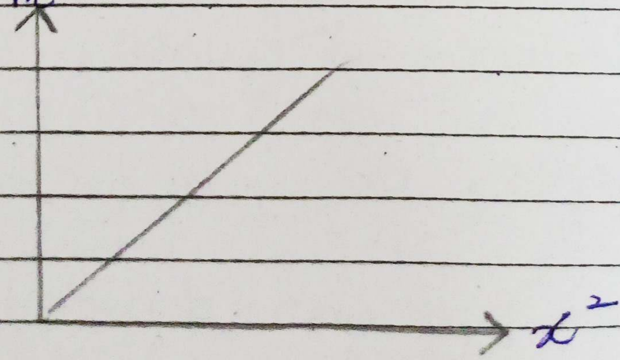


(d):- Parabolic Curve



↓

into linear curve



Plotting Of Curve On Simple /Plane Graph Paper

Rules

- It should have a tittle
- Labelling of both axis with quantities, units & multiplier
- Ranges & scaling of data
- 75% space must be covered by the curve
- Point should be plotted accurately & clearly
- If more than one curve are plotted than points must be differentiated
- Not necessary to start with zero value.

Conversion Of Non-Linear To Linear Curve

Rules

- by plotting quantities with exponent
- by using log or exponent of quantities
- by using semi-log or log to log graph paper
- curve on graph paper which donot have

constant

- slope(N
constant
the value
y-interce

Examp

Consider
If we dra
a non-lin
curve we
to linear e

Taking

where,

y

constant

↳ slope (N.L. curve) can be easily converted into constant slope and we can easily determine the value of constant from the linear curve by y-intercept and slope.

Example 1

Consider the radioactive decay equation $N = N_0 e^{-\lambda t}$
If we draw the graph b/w N and t we will find a non-linear curve. If we want to plot a linear curve we should convert the exponential equation to linear eq. by taking log on both sides

i.e $N = N_0 e^{-\lambda t}$

Taking log on both side;

$$\log N = \log(N_0 e^{-\lambda t})$$

$$\log N = \log N_0 + (-\lambda t) \log e$$

$$\log N = \log N_0 - \lambda t \ (1)$$

$$\log N = -\lambda t + \log N_0$$

where,

$y = \log N$; $mx = -\lambda t$; $c = \log N_0$

$$\underbrace{\log N}_{y} = \underbrace{-\lambda t}_{mx} + \underbrace{\log N_0}_{c}$$

where,

$$y = \log N$$
$$mx = -\lambda t$$
$$c = \log N$$

Thus, if we plot a curve b/w $\log N$ and t we will get a linear curve by determining the slope and y-intercept ~~bt~~ we can determine the value of decay constant (λ)

Example 2

If we draw the graph b/w T and $\sqrt{L}$, we will get a - linear curve. TO achieve a linear graph we will convert a non-linear equation into linear equation by taking log on both sides.

Date

Thus, if we plot a curve b/w $\ln N$ and t we will get a linear curve by determining the slope and y-intercept we can determine the value of decay constant 'λ'

Find y, mx, c of the following

i $V = IR$ ii $K.E = h\nu + \phi_0$ iii $V_f = V_i + at$

ii) $K.E = h\nu + \phi_0$

$$eV_0 = h\nu + \phi_0$$

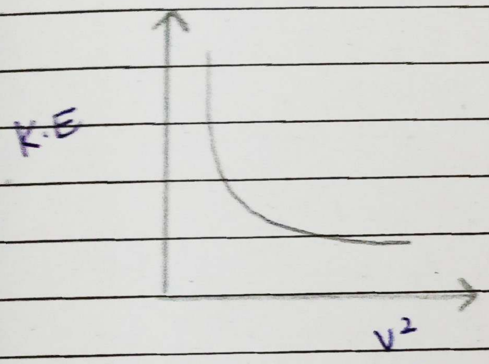
$$V_0 = \frac{h\nu}{e} + \frac{\phi_0}{e}$$

where;

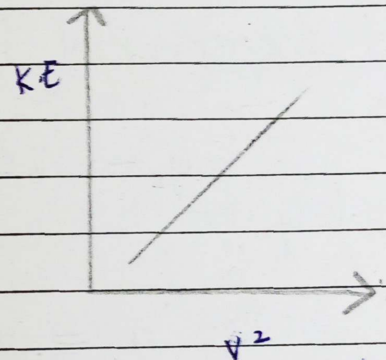
$y = V_0$, $mx = \frac{h\nu}{e}$, $c = \frac{\phi_0}{e}$

Example 3

Consider the equation of $K.E = \frac{1}{2} mv^2$
In this equation their is a direct relation b/w K.E and v^2. If we draw the graph b/w K.E and v^2 we will get a non-linear graph to convert this non-linear graph linear graph. we will consider the K.E on y-axis and v^2 on x-axis bcuz K.E increases with v^2



non-linear graph



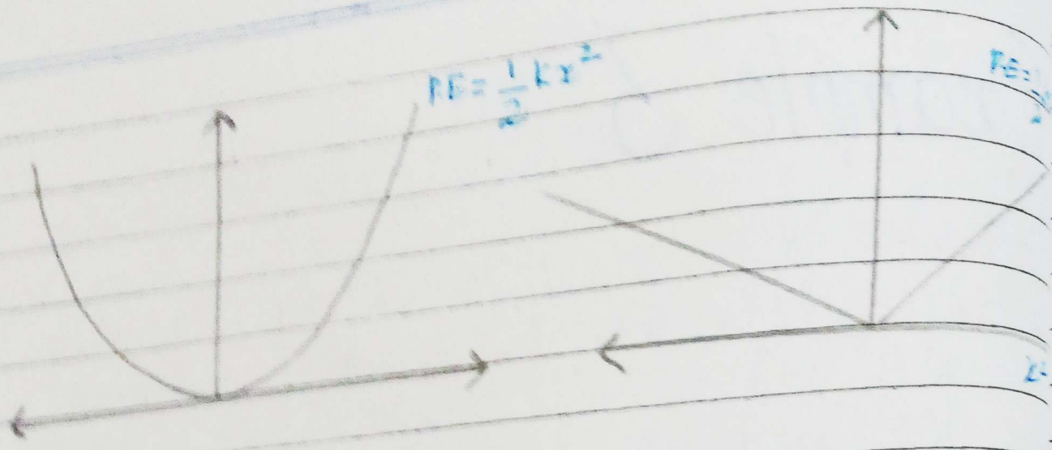
linear graph

Methods Of Converting Non-Linear Curve Into Linear Curve

Method # 1

Plotting Quantities With Power / Function

It can be made into linear by plotting a graph b/w P.E and x^2.
N.L.C don't have constant slope



Method #2

Plotting Quantities After Taking It's Log

— For Exponential Function

$$y = ae^{bx}$$

Taking log on b.s

$$\log y = \log(ae^{bx})$$

$$\log y = \log a + bx \log e$$

$$\log y = \log a + bx$$

$$\log y = bx + \log a$$

where,

$y = \log y$, $mx = b$, $c = \log a$

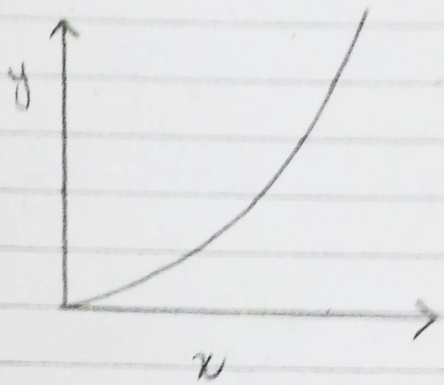
$z = bx + c$

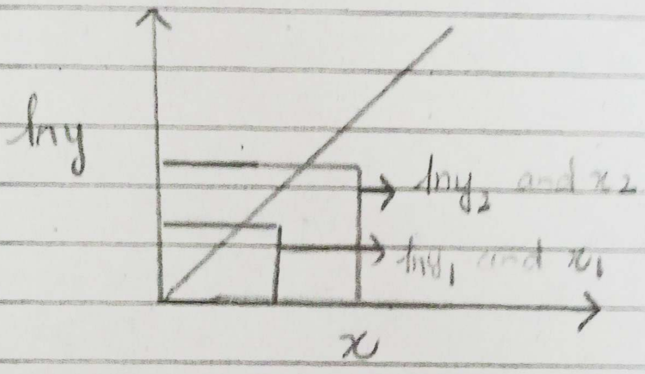
$$\text{slope} = \frac{z_2 - z_1}{x_2 - x_1} = b$$

$$b = \frac{\ln y_2 - \ln y_1}{x_2 - x_1}$$

$c = \ln a$

$a = e^c$





— For Polynomial Function

$$y = ax^b$$

Taking log on b.s

$$\log y = \log(ax^b)$$

$$\log y = \log a + b \log x$$

$$\log y = b \log x + \log a$$

where,

$$z = \log y$$

$$c = \log a$$

$$t = \log x$$

$$z = bt + c$$

$$\text{slope} = b = \frac{z_2 - z_1}{t_2 - t_1}$$

$$b = \frac{\log y_2 - \log y_1}{\log x_2 - \log x_1}$$

$$c = \log a \quad \Rightarrow \quad a = e^c$$

$$\ln = \log_c^a$$

Tak exp on bs

$$e(c) = e(\log a)$$

Log + e (cancle)

$$e(c) = a$$

$$a = e^c$$

Note

Plotting Of Graphs in PDF (Assignment)

(But Not Come in Paper)